

Hosting & Infrastructure Documentation

1. Overview

Our organizational project is currently hosted and managed using **Render** services under a single account. The infrastructure combines:

- **Render Web Services** for application hosting
- **Render Redis** for caching and session management
- **Supabase** for database and authentication

This document outlines the hosting plan, resource limits, and key considerations.

2. Hosting Platform: Render

- **Service Type:** Render Web Services
 - **Current Plan:** Free/Starter (under personal account)
 - **Deployment:** Automatic from GitHub repository (continuous deployment enabled)
 - **Region:** Default Render region (US-based)
 - **Limits:**
 - **CPU & Memory:** 512 MB RAM, 0.5 CPU (starter tier)
 - **Bandwidth:** Shared, limited under free tier
 - **Custom Domains:** Supported (requires DNS setup)
 - **SSL:** Auto-provisioned via Let's Encrypt
 - **Scaling:** Manual; autoscaling not available on free tier
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3. Caching Layer: Render Redis

- **Service Type:** Managed Redis instance
 - **Current Plan:** Free tier
 - **Limits:**
 - **Memory:** 25 MB max
 - **Connections:** Limited concurrent connections (approx. 30–50)
 - **Persistence:** No persistence enabled on free tier (volatile storage)
 - **Use Cases:** Session storage, caching, lightweight pub/sub
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4. Database: Supabase

- **Service Type:** Managed PostgreSQL + Storage
- **Current Plan:** Free tier
- **Limits:**
 - **Database Size:** 500 MB
 - **Row Limit:** ~500,000 rows
 - **API Requests:** 50,000 requests/month

- **Auth Users:** 50,000 users
 - **Storage:** 1 GB file storage
 - **Realtime:** Limited concurrent connections
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5. Current Constraints

- All services are running on **free tiers**, which impose strict limits on memory, storage, and concurrent connections.
- **No guaranteed uptime or SLA** — suitable for development and testing, but not production-critical workloads.
- **Scaling limitations:** Manual upgrades required to move beyond free tier limits.